328612(28)

CSVTUonline.com

B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(ET & T Engg. Branch)

ADVANCED ELECTRONIC CIRCUITS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: All questions are compulsory. Part (a) of each question is compulsory & attempt any two parts from (b), (c) & (d) parts of each questions.*

CSVTUonline.com

Unit-I

1. (a) What do you mean by weighted resistor? 2

(b) Explain the basic principle of successive

approximation with the help of suitable converter block diagram. 7

(c) Explain Invested R-2R ladder type D to A converter and derive expression for analog output signal. 7

(d) Draw and explain the basic circuit of sample and hold circuit. 7

CSVTUonline.com

Unit-II

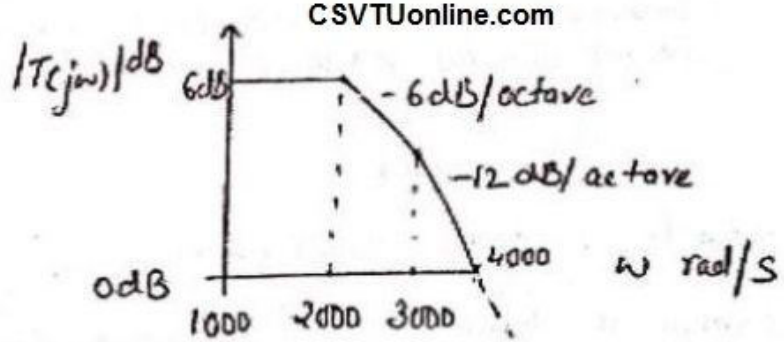
2. (a) What do you mean by bilinear transfer function? 2

(b) Derive an expression for low pass filter using 3 op-amp Biquad circuit. Also describe Tuning algorithm. 7

(c) Prove that Sallen & Key circuit gives the low pass filter response. Also give its transfer function. 7

(d) Design an amplifier filter from Bode plot & give schematic & element values. 7

CSVTUonline.com



Unit-III

3. (a) Explain RC-CR transformation. 2

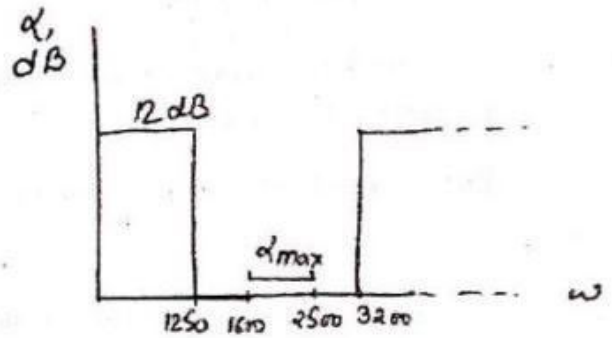
CSVTUonline.com

(b) Find the sensitivity of the Sallen Key circuit. 7

(c) The specification of 12 dB of attenuation is required in 2-stop band. It is prescribed that filter which satisfy these specification have only two op-amps. 7

(i) Determine value of α_{max} by Butterworth & Chebyshev.

(ii) For each of these find 2-half power frequency.



(d) Explain Delyiannis Friend Circuit. 7

CSVTUonline.com

Unit-IV

4. (a) State basic principle of operation of PLL with block diagram. 2

(b) Derive an expression for Lock-in range for phase Locked loop. 7

CSVTUonline.com

(c) Determine the free running frequency, the lock range and capture range for the circuit of PLL NE/565 IC, $V = 12$ V, $R_1 = 15\ k\Omega$, $C_1 = 0{\cdot}01\ \mu F$ & $C_2 = 10\ \mu F$. 7

(d) Draw the complete block diagram of voltage controlled oscillator (VCO) and derive an expression for frequency of oscillation. 7

Unit-V

5. (a) Define analog multiplier with symbol. 2

(b) Explain how monolithic multiplier (RC4200) is used as a divider and square rooting circuit. 7

(c) Explain different characteristics of analog multiplier with detail. 7

CSVTUonline.com

(d) Explain the basic method of performing analog multiplication. (Any two) 7